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Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: _____ **Section:** _____

Project Group ID: _____

ANALYSIS OF BIG DATA RESEARCH HOTSPOTS BASED ON KEYWORD CO-OCCURRENCE

The rapid growth of Big Data has led to a significant increase in research publications across various domains, including data mining, machine learning, artificial intelligence, cloud computing, data privacy, healthcare, and Industry 4.0. Due to the large volume and diversity of available research literature, it is difficult to manually identify major research themes, relationships among research topics, and emerging areas. Therefore, this project, **"Analysis of Big Data Research Hotspots Based on Keyword Co-occurrence,"** aims to provide a systematic and data-driven approach for identifying major research hotspots, thematic clusters, and emerging trends in Big Data research. The project is motivated by the need to simplify the analysis of large-scale research literature and provide useful insights for researchers in identifying important and future research directions. The proposed methodology involves collecting relevant research publications from academic databases, pre-processing the collected data, extracting and standardizing keywords, and analyzing the frequency and co-occurrence of keywords. A keyword co-occurrence matrix is constructed to identify relationships between frequently occurring keywords, followed by network analysis and clustering techniques to group related keywords into meaningful research themes. The expected outcomes include the identification of prominent Big Data research hotspots, relationships between major research topics, thematic clusters, and emerging areas of interest. The project also contributes to the particularly **SDG 4: Quality Education** by supporting data-driven research and knowledge development, **SDG 8: Decent Work and Economic Growth** by identifying technology and innovation trends that can support future employment and economic development, and **SDG 9: Industry, Innovation and Infrastructure** by highlighting emerging technologies and research directions that contribute to digital innovation.

Keywords: Big Data, Data Mining, Keyword Co-occurrence, Text Mining, Research Hotspots, Bibliometric Analysis.

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor: Mr. Asif Ahmed

Signature: _____ **Date**: _____

Coordinator **Head of the Department**